

Analytics Dashboard For Titanic Dataset

Introduction

The purpose of this project is to develop an interactive Business Intelligence (BI) dashboard for analyzing the Titanic passenger dataset and predicting survival outcomes using machine learning.

The dashboard allows users to explore passenger demographics, visualize survival trends, and generate predictions through an easy-to-use web interface.

The project is built using Python, Streamlit, Pandas, Plotly, and Scikit-learn.

Objectives

The main objectives of this project are:

- To analyze Titanic passenger data through interactive visualizations.
- To create meaningful features for better insights.
- To provide dynamic filtering options for data exploration.
- To build a machine learning model for survival prediction.
- To combine analytics and predictive modeling in a single dashboard.

Dataset and Feature Engineering

The project uses the cleaned Titanic dataset containing passenger information such as age, gender, ticket class, fare, cabin details, and survival status.

To improve analysis, several new features are created:

- Family Size = Number of siblings/spouses + number of parents/children.
- Age Brackets to group passengers into categories such as Child, Teenager, Adult, and Senior.
- Deck Information extracted from cabin codes.
- Passenger Titles extracted from names (Mr., Mrs., Miss, etc.).

These features help reveal patterns that are not immediately visible in the original dataset.

Dashboard Features

The dashboard provides interactive filters for:

- Passenger name search
- Gender selection

- Passenger class selection
- Deck-level filtering

Based on the selected filters, the dashboard displays key metrics including:

- Total passengers
- Survival rate
- Average ticket fare
- Most common passenger title

Several interactive visualizations are included:

- Survival flow analysis by class and gender
- Survival heatmap by age group and passenger class
- Fare distribution comparison
- Deck-wise survival analysis

These visualizations help users understand how different factors influence survival chances.

Machine Learning Model

A Random Forest Classifier is implemented to predict passenger survival.

The model uses the following features:

- Passenger class
- Gender
- Age
- Fare
- Family size
- Embarkation port

Users can enter hypothetical passenger details through the dashboard, and the model predicts whether the passenger survives along with a confidence score.

The dashboard also displays feature importance scores to show which factors have the greatest impact on predictions.

Conclusion

This project successfully combines data visualization and machine learning within a single interactive dashboard.

Users can explore historical Titanic data, identify important survival patterns, and test custom passenger profiles using predictive analytics.

The project demonstrates practical applications of Business Intelligence, feature engineering, data visualization, and machine learning while providing an engaging and user-friendly analytical experience.